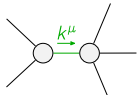


$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}_{\dagger} & \begin{array}{|c|c|} \hline -\frac{3k^2{}^{(4)}\kappa_1}{2} & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#1}{}_{\alpha} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -k^2{}^{(4)}\kappa_3 & -\frac{k^2{}^{(4)}\kappa_3}{\sqrt{2}} \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{k^2{}^{(4)}\kappa_3}{\sqrt{2}} & -\frac{k^2{}^{(4)}\kappa_3}{2} \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\end{array}$$

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}_{\dagger} & \begin{array}{|c|c|} \hline -\frac{2}{3k^2{}^{(4)}\kappa_1} & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#1}{}_{\alpha} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{4}{9k^2{}^{(4)}\kappa_3} & -\frac{2\sqrt{2}}{9k^2{}^{(4)}\kappa_3} \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\frac{2\sqrt{2}}{9k^2{}^{(4)}\kappa_3} & -\frac{2}{9k^2{}^{(4)}\kappa_3} \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|} \hline 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\end{array}$$

### Lagrangian

$$\kappa_1^{(4)}\partial_\beta\mathcal{K}_{\chi\delta}^\delta\partial^\chi\mathcal{K}_\alpha^{\alpha\beta} + \kappa_3^{(4)}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\alpha\chi}^\delta - 2\kappa_3^{(4)}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}^\delta - \kappa_3^{(4)}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\chi\alpha}^\delta - \frac{1}{2}\kappa_3^{(4)}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}_{\beta\chi}^\delta$$

| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             |
| $\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$                                      | 1                                                             | $ \begin{aligned} &\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \\ &\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} \end{aligned} $                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                             |
| $\mathcal{J}_{1^+}^{\#2\alpha} == 0$                                               | 3                                                             | $\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi} == 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
| $\mathcal{J}_{1^+}^{\#1\alpha} == 0$                                               | 3                                                             | $\partial_\chi\partial^\alpha\mathcal{J}^{\beta\chi}{}_\chi + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                             |
| $\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$         | 3                                                             | $3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                             |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$                                      | 5                                                             | $ \begin{aligned} &3\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \\ &4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_\epsilon{}^\epsilon + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha}{}_\delta == \\ &3\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \\ &2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_\epsilon{}^\epsilon + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta}{}_\delta \end{aligned} $ |                             |
| $\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$                                          | 5                                                             | $3\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\eta^{\alpha\beta}\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_\chi{}^\delta == 2\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_\chi{}^\delta + 3(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |
| Total # constraint(s):                                                             | 20                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                             |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Residue                     |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $-\frac{1}{\kappa_3^{(4)}}$ |

Resolved unitarity condition(s):

$$\kappa_3^{(4)} < 0$$